

MATH 2123

Mathematics III - 2020

Beginner Guide + Full Step-by-Step Solutions

Khulna University of Engineering & Technology
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How to use this guide

For every main question, first learn the topic and minimum basics, then follow the repeatable procedure, and finally study the full solution. The 2020 paper has 6 main question sets (not 8), because it is a 120-mark paper.

Beginner promise

You do not need to be an expert before starting. The goal is to recognize the question type, choose the correct formula, and execute the steps safely in an exam.

Paper map - what each set is testing

Set	Main topic	What to learn
1	Vectors in 3D	Coplanarity, triple product, vector triple-product identity, angle between vectors
2	Directional derivatives + line integral	Recovering a gradient from directional derivatives; work along a parametric curve
3	Matrices + multivariable calculus	Eigenpairs, consistency of linear systems, directional derivative of a scalar field
4	Laplace transform	Piecewise/unit-step transform, inverse transform, improper integral using Laplace properties
5	Laplace applications	Volterra convolution integral equation; fourth-order differential equation
6	Matrix algebra	Adjoint and inverse; rank depending on a parameter

High-yield formulas

Coplanarity: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$. Directional derivative: $D_{\mathbf{u}} f = \text{grad}(f) \cdot \mathbf{u}_{\text{hat}}$. Work: $W = \int \mathbf{F} \cdot d\mathbf{r}$. Eigenvalues: $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$. Laplace shift: $L\{u(t-a)g(t-a)\} = e^{-as}G(s)$. Inverse matrix: $\mathbf{A}^{-1} = \text{adj}(\mathbf{A})/\det(\mathbf{A})$. Rank 3 for a 3×3 matrix when $\det(\mathbf{A})$ is nonzero.

Set 1 - 3D Vectors

Topic

Coplanarity of three vectors, vector triple products, and angle between vectors.

Basics you need

- Three vectors are coplanar exactly when their scalar triple product is zero: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.
- The determinant made from the vector components equals the scalar triple product.
- Vector triple-product identity: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$.
- If $\mathbf{u} \cdot \mathbf{v} = 0$ and both are nonzero, the angle between them is 90 degrees.

1(a) Find lambda so the three vectors are not coplanar

$$\mathbf{v}_1 = (\lambda, -1, 1), \mathbf{v}_2 = (1, 2, -3), \mathbf{v}_3 = (3, -4, 5)$$

Problem-solving procedure

1. Form the 3×3 determinant using the components.
2. Set determinant not equal to zero because the question says not coplanar.
3. Solve the resulting inequality for lambda.

$$\Delta = \begin{vmatrix} \lambda & -1 & 1 \\ 1 & 2 & -3 \\ 3 & -4 & 5 \end{vmatrix}$$

$$\Delta = -2(\lambda - 2)$$

$$\text{Not coplanar} \Rightarrow \Delta \neq 0 \Rightarrow \lambda \neq 2.$$

Final answer

lambda can be any real number except 2.

Common mistake

Do not set the determinant equal to zero here. Zero means coplanar; the question asks for not coplanar.

1(b) Vector triple-product identity and angle

$$\mathbf{a} = (2, 3, 4), \mathbf{b} = (1, 1, -1), \mathbf{c} = (1, -1, 1)$$

Problem-solving procedure

1. Compute $\mathbf{b} \times \mathbf{c}$.
2. Compute $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.
3. Separately evaluate $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ to verify the identity.
4. Use dot product to find the angle between $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ and $\mathbf{b} \times \mathbf{c}$.

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} = (0, -2, -2).$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (2, 3, 4) \times (0, -2, -2) = (2, 4, -4).$$

$$\mathbf{a} \cdot \mathbf{c} = 2(1) + 3(-1) + 4(1) = 3, \quad \mathbf{a} \cdot \mathbf{b} = 2 + 3 - 4 = 1.$$

$$(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = 3(1, 1, -1) - (1)(-1, 1, 1) = (2, 4, -4).$$

$$(2, 4, -4) \cdot (0, -2, -2) = 0, \text{ so the two nonzero vectors are perpendicular.}$$

Final answer

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (2, 4, -4) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}; \text{ angle} = 90 \text{ degrees } (\pi/2).$$

Exam tip

Whenever you see $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$, remember the BAC-CAB identity. It can save several lines of cross-product calculation.

Set 2 - Directional Derivatives and Work

Topic

Using directional-derivative data to recover the gradient, then evaluating work along a specified curve.

Basics you need

- Directional derivative in direction $\mathbf{v}$ uses the unit vector $\mathbf{u} = \mathbf{v}/|\mathbf{v}|$: $D_{\mathbf{u}} f = \text{grad}(f) \cdot \mathbf{u}$.
- For $f(x,y)$, $\text{grad}(f) = (f_x, f_y)$. Two independent directional-derivative facts can determine f_x and f_y at a point.
- Work along a curve C is $W = \int_C \mathbf{F} \cdot d\mathbf{r}$. Parameterize the curve first.

2(a) Recover the gradient at P(1,2)

Let $\text{grad}(f)(1,2) = (p,q)$.

Problem-solving procedure

- Convert each stated direction into a unit vector.
- Write a dot-product equation for each known directional derivative.
- Solve for p and q .
- Dot the recovered gradient with the unit vector of the requested direction.

Direction $\mathbf{i} + \mathbf{j}$ has unit vector $(1,1)/\sqrt{2}$.

$$(p+q)/\sqrt{2} = 2\sqrt{2} \Rightarrow p+q=4.$$

Direction $-2\mathbf{j}$ has the same unit direction as $-\mathbf{j} = (0,-1)$.

$$(p,q) \cdot (0,-1) = -q = -3 \Rightarrow q=3, \text{ hence } p=1.$$

Requested direction $-\mathbf{i} - 2\mathbf{j}$ has unit vector $(-1,-2)/\sqrt{5}$.

$$D = (1,3) \cdot (-1,-2)/\sqrt{5} = (-1-6)/\sqrt{5} = -7/\sqrt{5}.$$

Final answer

Directional derivative in the direction $-\mathbf{i} - 2\mathbf{j}$ is $-7/\sqrt{5}$.

Common mistake

A direction vector is normally normalized before taking a directional derivative. The length of $-2\mathbf{j}$ does not make the derivative twice as large.

2(b) Work done along the curve

$$\mathbf{F} = 3x^2 \mathbf{i} + (2xz - y) \mathbf{j} + z \mathbf{k}$$

$$\text{Curve: } x^2 = 4y, \quad 3x^3 = 8z, \quad x: 0 \text{ to } 2.$$

Problem-solving procedure

- Use x itself as the parameter.
- Express y and z as functions of x .
- Compute $d\mathbf{r} = (dx, dy, dz)$.
- Substitute into $\mathbf{F} \cdot d\mathbf{r}$ and integrate from $x=0$ to $x=2$.

$$y = x^2/4, \quad z = 3x^3/8.$$

$$d\mathbf{r} = (1, x/2, 9x^2/8) dx.$$

$$\mathbf{F} = (3x^2, 3x^4/4 - x^2/4, 3x^3/8).$$

$$\mathbf{F} \cdot d\mathbf{r} = 3x^2 + (3x^4/4 - x^2/4)(x/2) + (3x^3/8)(9x^2/8).$$

$$= 3x^2 - x^3/8 + 51x^5/64.$$

$$\begin{aligned} W &= \int_0^2 [3x^2 - x^3/8 + 51x^5/64] \, dx \\ &= [x^3 - x^4/32 + 51x^6/384]_0^2 = 16. \end{aligned}$$

Final answer

Work done = 16 units.

Exam tip

When the curve is given as equations involving x , choosing x as the parameter is usually the fastest route.

Set 3 - Eigenvalues, Consistency, and Gradient

Topic

Three common exam patterns: eigenpairs of a small matrix, parameter-dependent consistency, and a standard directional derivative.

3(a) Eigenvalues and eigenvectors

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Basics you need

- Eigenvalues satisfy $\det(A - \lambda I) = 0$.
- For each eigenvalue λ , solve $(A - \lambda I)x = 0$ for a nonzero eigenvector.

Problem-solving procedure

1. Form $\det(A - \lambda I)$.
2. Solve the characteristic equation.
3. Back-substitute each eigenvalue to obtain an eigenvector.

$$\begin{aligned} \det\left(\begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}\right) &= (1-\lambda)(3-\lambda) - 8 \\ &= \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1). \end{aligned}$$

Therefore $\lambda = 5$ or $\lambda = -1$.

For $\lambda = 5$: $-4x + 4y = 0 \Rightarrow x = y$. Choose $v = (1, 1)$.

For $\lambda = -1$: $2x + 4y = 0 \Rightarrow x = -2y$. Choose $v = (-2, 1)$.

Final answer

Eigenpairs: $\lambda = 5$ with v proportional to $(1, 1)$; $\lambda = -1$ with v proportional to $(-2, 1)$.

3(b) Consistency of a parameter-dependent system

Augmented matrix: $\begin{bmatrix} a & 1 & | & 1 \\ 2 & a-1 & | & 1 \end{bmatrix}$

$$\text{Coefficient determinant} = a(a-1) - 2 = (a-2)(a+1).$$

Problem-solving procedure

1. If determinant is nonzero, the system has one unique solution.
2. If determinant is zero, inspect the two equations to decide between infinitely many and no solution.
 - If $a \neq 2, -1$: determinant $\neq 0 \Rightarrow$ unique solution.
 - Indeed $x = y = 1/(a+1)$ for these ordinary values.
 - If $a = 2$: both equations become $2x + y = 1 \Rightarrow$ infinitely many solutions.
 - If $a = -1$: equations are $-x + y = 1$ and $2x - 2y = 1$. The left sides are multiples but the right sides are not $\Rightarrow$ inconsistent.

Final answer

Consistent for all a except -1 . Unique for $a \neq -1, 2$; infinitely many for $a = 2$; no solution for $a = -1$.

3(c) Directional derivative of ϕ

$\phi = x^2 y z + 4xz^2$, at $(1, -2, -1)$, direction $v = (2, -1, -2)$.

Problem-solving procedure

1. Find $\text{grad}(\phi)$.
2. Evaluate the gradient at the point.

3. Normalize the direction vector.

4. Take the dot product.

$$\phi_x = 2xyz + 4z^2; \quad \phi_y = x^2z; \quad \phi_z = x^2y + 8xz.$$

$$\text{grad}(\phi)(1, -2, -1) = (8, -1, -10).$$

$$|v| = 3, \text{ so } u = (2, -1, -2)/3.$$

$$D_u \phi = (8, -1, -10) \cdot (2, -1, -2)/3 = (16 + 1 + 20)/3 = 37/3.$$

Final answer

Directional derivative = $37/3$.

Set 4 - Laplace Transform Fundamentals

Topic

Piecewise functions, inverse Laplace by partial fractions, and evaluating an improper integral using a transform derivative property.

Basics you need

- $L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$.
- Unit step: $u(t-a)=0$ before a and 1 after a .
- $L\{u(t-a)g(t-a)\} = e^{-as}G(s)$.
- $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} [F(s)]$.

4(a) Laplace transform of the piecewise function

$f(t)=\sin t, 0 \leq t < \pi/4; f(t)=\sin t + \cos(t-\pi/4), t \geq \pi/4$.

Problem-solving procedure

1. Recognize the extra term that switches on at $t=\pi/4$.
2. Rewrite with a unit-step function.
3. Apply the second shifting theorem.

Let $a=\pi/4$. Then $f(t)=\sin t + u(t-a) \cos(t-a)$.

$L\{\sin t\}=1/(s^2+1), L\{u(t-a)\cos(t-a)\}=e^{-as} s/(s^2+1)$.

Therefore $F(s) = [1 + s e^{-\pi s/4}]/(s^2+1)$.

Final answer

$L\{f(t)\} = (1 + s e^{-\pi s/4})/(s^2+1)$.

4(b) Inverse Laplace transform

Find $L^{-1}\{1/[(s+2)^4(s-2)]\}$.

Problem-solving procedure

1. Use partial fractions because the denominator has repeated linear factors.
2. Convert each standard term using $L^{-1}\{1/(s+a)^n\}=t^{n-1}e^{-at}/(n-1)!$ and $L^{-1}\{1/(s-a)\}=e^{at}$.

$$1/[(s+2)^4(s-2)] = -1/[256(s+2)] - 1/[64(s+2)^2]$$

$$-1/[16(s+2)^3] - 1/[4(s+2)^4] + 1/[256(s-2)].$$

$$f(t) = -e^{-(2t)}/256 - t e^{-(2t)}/64 - t^2 e^{-(2t)}/32 - t^3 e^{-(2t)}/24 + e^{(2t)}/256.$$

Final answer

$f(t) = (e^{(2t)}-e^{-(2t)})/256 - (t/64 + t^2/32 + t^3/24)e^{-(2t)}$.

4(c) Evaluate $\int_0^{\infty} t^3 e^{-t} \sin t dt$

Problem-solving procedure

1. Start from $F(s)=L\{\sin t\}=1/(s^2+1)$.
2. Use $L\{t^3 \sin t\} = -d^3F/ds^3$.
3. The given integral equals $L\{t^3 \sin t\}$ evaluated at $s=1$.

$$d^3F/ds^3 = -24s(s-1)(s+1)/(s^2+1)^4.$$

$$\text{Integral} = -[d^3F/ds^3] \text{ at } s=1 = 0.$$

Final answer

The value of the integral is 0.

Exam tip

A zero answer is possible here: the oscillating sine produces positive and negative contributions that cancel after weighting.

Set 5 - Laplace Transform Applications

Topic

Solving a Volterra convolution integral equation and a fourth-order ODE with mixed conditions.

5(a) Integral equation by Laplace transform

$$F(t) = 1 + 2 \int_0^t F(t-x)e^{-2x} dx.$$

Basics you need

- The integral is a convolution: $(F * e^{-2t})(t)$.
- Laplace of a convolution is the product of the transforms.

Problem-solving procedure

1. Take Laplace transforms on both sides.
2. Replace the convolution by a product.
3. Solve algebraically for the transform.
4. Invert.

Let $L\{F(t)\} = Y(s)$. Then

$$Y = 1/s + 2 Y/(s+2).$$

$$Y[1 - 2/(s+2)] = 1/s \Rightarrow Y[s/(s+2)] = 1/s.$$

$$Y = (s+2)/s^2 = 1/s + 2/s^2.$$

Therefore $F(t)=1+2t$.

Final answer

$$F(t) = 1 + 2t.$$

5(b) Fourth-order differential equation

$$y'''' - 16y = 30 \sin t$$

Conditions: $y(0)=0$, $y'(0)=2$, $y''(\pi)=0$, $y'''(\pi)=-18$.

Basics you need

- For a fourth derivative, $L\{y^{(4)}\}=s^4Y-s^3y(0)-s^2y'(0)-sy''(0)-y'''(0)$.
- Because two conditions are at $t=\pi$, temporarily call $y''(0)=A$ and $y'''(0)=B$, then use the $t=\pi$ conditions after inversion.

Problem-solving procedure

1. Introduce $A=y''(0)$, $B=y'''(0)$.
2. Transform the ODE.
3. Partial-fraction and invert to get $y(t)$ in terms of A, B .
4. Use $y''(\pi)=0$ and $y'''(\pi)=-18$ to determine A, B .

$$(s^4-16)Y = 30/(s^2+1) + 2s^2 + As + B.$$

After partial fractions:

$$Y = -(As+B-18)/[8(s^2+4)] - 2/(s^2+1) \\ + (2A-B-14)/[32(s+2)] + (2A+B+14)/[32(s-2)].$$

Hence, for $t \geq 0$,

$$y = -(A/8)\cos 2t - (B-18)/16 \sin 2t - 2\sin t$$

$$+(2A-B-14)e^{(-2t)/32} + (2A+B+14)e^{(2t)/32}.$$

Using $y''(\pi)=0$ and $y'''(\pi)=-18$ gives $A=0$ and $B=-14$.

Substituting these values cancels the exponential and cosine terms:

$$y(t) = 2 \sin 2t - 2 \sin t.$$

Final answer

$$y(t) = 2 \sin(2t) - 2 \sin(t).$$

Common mistake

Do not treat $y''(\pi)$ and $y'''(\pi)$ as initial values at $t=0$. Introduce the unknown initial derivatives first, then enforce the conditions at π .

Set 6 - Adjoint, Inverse, and Rank

Topic

Computing an inverse from cofactors and classifying matrix rank as a parameter changes.

6(a) Adjoint and inverse

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 7 & 8 \\ 4 & 10 & 12 \end{bmatrix}$$

Basics you need

- The cofactor $C_{ij} = (-1)^{i+j}$ times the minor determinant.
- $\text{adj}(A)$ is the transpose of the cofactor matrix.
- If $\det(A) \neq 0$, $A^{-1} = \text{adj}(A)/\det(A)$.

Problem-solving procedure

1. Compute $\det(A)$.
2. Compute all cofactors and transpose the cofactor matrix.
3. Divide the adjoint by the determinant.

$$\det(A) = 36.$$

$$\text{adj}(A) = \begin{bmatrix} 4 & 16 & -12 \\ -28 & -4 & 12 \\ 22 & -2 & -3 \end{bmatrix}.$$

$$\begin{aligned} A^{-1} &= (1/36) \text{adj}(A) \\ &= \begin{bmatrix} 1/9 & 4/9 & -1/3 \\ -7/9 & -1/9 & 1/3 \\ 11/18 & -1/18 & -1/12 \end{bmatrix}. \end{aligned}$$

Final answer

The inverse exists because $\det(A) = 36 \neq 0$; A^{-1} is the matrix shown above.

6(b) Rank of a real matrix depending on a

$$M = \begin{bmatrix} a & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-a \end{bmatrix}.$$

Problem-solving procedure

1. Compute the 3×3 determinant.
2. If it is nonzero, $\text{rank} = 3$.
3. For parameter values making the determinant zero, check a 2×2 minor.

$$\det(M) = -(a-2)(a+1).$$

Therefore, if $a \neq 2, -1$, $\det(M) \neq 0$ and $\text{rank}(M) = 3$.

At $a = 2$, the minor $\begin{vmatrix} a & 1 \\ 1 & 1 \end{vmatrix} = a - 1 = 1 \neq 0$, so $\text{rank} = 2$.

At $a = -1$, the same minor equals $-2 \neq 0$, so $\text{rank} = 2$.

Final answer

$\text{rank}(M) = 3$ for $a \neq 2, -1$; $\text{rank}(M) = 2$ for $a = 2$ or $a = -1$.

Rapid Revision - Procedures to Memorize

Question pattern	Fast procedure
Coplanarity	Put the three vectors into a determinant. $\det=0 \rightarrow$ coplanar; $\det \neq 0 \rightarrow$ not coplanar.
Triple product identity	$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$.
Directional derivative	Normalize direction $\mathbf{v}$, then $\text{grad}(\mathbf{f}) \cdot \mathbf{v}_{\text{hat}}$.
Line integral/work	Parameterize $C \rightarrow$ compute $d\mathbf{r} \rightarrow$ substitute into $\mathbf{F} \rightarrow$ integrate $\mathbf{F} \cdot d\mathbf{r}$.
Eigenpair	$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$, then solve $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = 0$.
Consistency	Nonzero determinant $\rightarrow$ unique. Zero determinant $\rightarrow$ compare ranks/equations.
Piecewise Laplace	Rewrite switching terms with $u(t-a)$, then use $e^{(-as)}$ shift.
Repeated inverse Laplace	Partial fractions; $1/(s+a)^n \rightarrow t^{(n-1)}e^{(-at)}/(n-1)!$.
Laplace integral property	$\int_0^\infty e^{(-st)} t^n f(t) dt = (-1)^n F^{(n)}(s)$.
Convolution equation	Transform convolution into a product, solve for transform, invert.
Inverse by adjoint	$\mathbf{A}^{-1} = \text{adj}(\mathbf{A})/\det(\mathbf{A})$, only if $\det(\mathbf{A}) \neq 0$.
Rank with parameter	Use determinant first, then test lower-order minors at exceptional parameter values.

Final Answers at a Glance

Part	Final result
1(a)	$\lambda = 2$.
1(b)	$a \times (b \times c) = (2, 4, -4)$; identity matches; angle with $b \times c = 90$ degrees.
2(a)	$-7/\sqrt{5}$.
2(b)	Work = 16.
3(a)	Eigenvalues 5 and -1; eigenvectors proportional to (1,1) and (-2,1).
3(b)	Consistent for $a \neq -1$. Unique for $a \neq -1, 2$; infinite at $a=2$; none at $a=-1$.
3(c)	$37/3$.
4(a)	$(1 + s e^{-\pi s/4})/(s^2 + 1)$.
4(b)	$(e^{2t} - e^{-2t})/256 - (t/64 + t^2/32 + t^3/24)e^{-2t}$.
4(c)	0.
5(a)	$F(t) = 1 + 2t$.
5(b)	$y(t) = 2 \sin(2t) - 2 \sin t$.
6(a)	$\det = 36$; inverse = $\begin{bmatrix} 1/9 & 4/9 & -1/3 \\ -7/9 & -1/9 & 1/3 \\ 11/18 & -1/18 & -1/12 \end{bmatrix}$.
6(b)	rank=3 if $a \neq 2, -1$; rank=2 if $a=2$ or -1 .

Last-minute exam strategy

First identify the pattern. Write the controlling formula before doing algebra. Normalize every directional vector. In line integrals, parameterize before substituting. In matrix questions, use the determinant to detect special parameter values. In Laplace questions, look for unit-step shifts, convolution, and repeated factors.